

Numerical, for Electrode Potential and Cells:

1. Calculate the electrode potential of Zinc, when the standard potential of Zinc electrode is -0.76 V and concentration of Zn^{2+} ion is 0.1M

$$E^0_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V},$$

$$[\text{Zn}^{2+}] = 0.1\text{M}$$

$$E_{\text{Zn}^{2+}/\text{Zn}} = E^0_{\text{Zn}^{2+}/\text{Zn}} + \frac{0.0591}{n} \log [\text{Zn}^{2+}]$$

$$= -0.76\text{V} + \frac{0.0591}{2} \log (0.1)$$

$$= -0.7895\text{V}$$

$$E_{\text{cell}} = E^0 - \frac{0.0591}{n} \log \frac{[P]}{[R]} \quad \text{Or} \quad E_{\text{cell}} = E^0 + \frac{0.0591}{n} \log \frac{[R]}{[P]}$$

$$E_{\text{Mn}^{n+}/\text{M}} = E^0_{\text{Mn}^{n+}/\text{M}} - \frac{0.0591}{n} \log \frac{1}{[\text{M}^{n+}]} \quad \text{Or} \quad E_{\text{Mn}^{n+}/\text{M}} = E^0_{\text{Mn}^{n+}/\text{M}} + \frac{0.0591}{n} \log [\text{M}^{n+}]$$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \frac{[\text{M}_2]}{[\text{M}_1]} \quad \text{M}_2 > \text{M}_1$$

2. Calculate at 25°C the electrode potential of $\text{Fe}^{2+}(0.1\text{M})/\text{Fe}$ given $E^0_{\text{Fe}^{2+}/\text{Fe}} = -0.44\text{V}$.

$$E^0_{\text{Fe}^{2+}/\text{Fe}} = -0.76\text{V},$$

$$[\text{Fe}^{2+}] = 0.1\text{M}$$

$$E_{\text{Fe}^{2+}/\text{Fe}} = E^0_{\text{Fe}^{2+}/\text{Fe}} + 0.0591 \log [\text{Fe}^{2+}]$$

$$= -0.44 + 0.0591 \log$$

$$= -0.44 + 0.0591 \log$$

$$= -0.44 - 0.0591 (1)$$

$$= -0.4695\text{V}$$

3. Calculate the reduction potential of copper when it is in contact with 5M CuSO_4 solution at 298K. The E^0 value of copper electrode is 0.34 V.

$$E_{\text{Cu}^{2+}/\text{Cu}} = E^0_{\text{Cu}^{2+}/\text{Cu}} + \frac{0.0591}{n} \log [\text{Cu}^{2+}]$$

$$E_{\text{Cu}^{2+}/\text{Cu}} = 0.34 + \frac{0.0591}{2} \log [5]$$

$$= 0.36\text{Volts}$$

4. Calculate the electrode potential of copper, its electrode potential at 25 °C is 0.292V when $[\text{Cu}^{2+}] = 0.015\text{M}$.

$$E_{\text{Cu}^{2+}/\text{Cu}} = E^0_{\text{Cu}^{2+}/\text{Cu}} + \frac{0.0591}{n} \log [\text{Cu}^{2+}]$$

$$E_{\text{Cu}^{2+}/\text{Cu}} = 0.292 + \frac{0.0591}{2} \log [0.015]$$

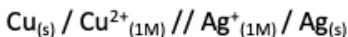
$$E_{\text{Cu}^{2+}/\text{Cu}} = 0.3498\text{V}$$

5. Calculate the emf of a cell containing Copper and silver electrode given that the electrode potentials of copper and silver are 0.34 V and 0.8V respectively.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = 0.8 - 0.34$$

$$E_{\text{cell}} = 0.46\text{V}$$

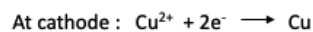


6. A galvanic cell formed by immersing Cd rod in 1M CdSO_4 and Cu rod in 1M CuSO_4 solution. The standard E^0 values of Cu and Cd electrodes are +0.34V and -0.40V respectively.

- How the cell is represented?
- Write the electrode reactions and cell reaction
- Calculate the standard emf of the cell at 298K.

a. Cell representation $\text{Cd}_{(s)} / \text{Cd}^{2+}_{(1M)} // \text{Cu}^{2+}_{(1M)} / \text{Cu}_{(s)}$

b. Electrode reactions are



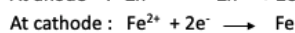
c. EMF of cell.

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} \\ &= E_{\text{Cu}} - E_{\text{Cd}} \\ &= 0.34 - (-0.4) \\ E_{\text{cell}} &= 0.74 \text{ Volts} \end{aligned}$$

7. Write the electrode reactions, cell reaction and calculate the emf of the cell at 298K for the cell formed by $\text{Fe}^{2+} / \text{Fe}$, $E^0 = -0.44$ and $\text{Zn}^{2+} / \text{Zn}$, $E^0 = -0.76\text{V}$ respectively.

a. Cell representation $\text{Zn} / \text{Zn}^{2+} // \text{Fe}^{2+} / \text{Fe}$

b. Electrode reactions are

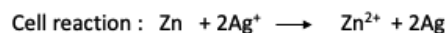
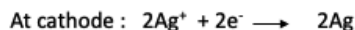


c. EMF of cell.

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} \\ &= -0.44 - (-0.76) \\ &= 0.32 \text{ V} \end{aligned}$$

8. Calculate the potential of Ag-Zn cell at 298K if the concentration of Ag^+ and Zn^{2+} are $5.2 \times 10^{-6}\text{M}$ and $1.3 \times 10^{-3}\text{M}$ respectively E^0 of the cell at 298K is 1.5V. Calculate the change in free energy ΔG for the reduction of 1 mole of Ag^+ . Given that 1 Faraday = $96.5\text{KJ V}^{-1}\text{mol}^{-1}$ (KJ per volt gram equivalent)

Electrode reactions



$$E_{\text{Cell}} = E^0_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$$

$$\begin{aligned} E_{\text{Cell}} &= 1.5 - \frac{0.0591}{2} \log \frac{[1.3 \times 10^{-3}]}{[5.2 \times 10^{-6}]^2} \\ &= 1.251 \text{ V.} \end{aligned}$$

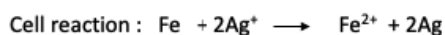
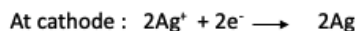
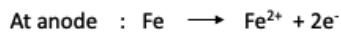
Change in free energy for the reduction 1 mole of Ag^+

$$\begin{aligned} \Delta G &= -nFE \\ &= -1 \times 96.5 \times 1.251 \\ \Delta G &= -120.7 \text{ KJ} \end{aligned}$$

9. An electrochemical cell consists of iron electrode dipped in 0.1M FeSO_4 and Silver electrode dipped in 0.05M AgNO_3 . Write the cell representation, cell reaction and calculate the electrode potential of the cell at 298 K. Given that the standard reduction potential of iron and silver electrodes are -0.44 and +0.80V respectively.

Cell representation $\text{Fe}_{(s)} / \text{Fe}^{2+}_{(0.1M)} // \text{Ag}^+_{(0.05M)} / \text{Ag}_{(s)}$

Electrode reactions are



EMF of cell.

$$\begin{aligned} E^0_{\text{cell}} &= E^0_{\text{Ag}^+/\text{Ag}} - E^0_{\text{Fe}/\text{Fe}^{2+}} \\ &= 0.80 - (-0.44) \\ E^0_{\text{cell}} &= 1.24 \text{ V} \end{aligned}$$

E_{cell}

$$E_{\text{Cell}} = E^{\circ}_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[\text{Fe}^{2+}]}{[\text{Ag}^+]^2}$$

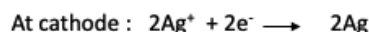
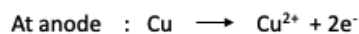
$$E_{\text{Cell}} = 1.24 - \frac{0.0591}{2} \log \frac{[0.1]}{[0.05]^2}$$

$$E_{\text{cell}} = 1.1927 \text{ V}$$

10. Write the electrode reaction and calculate the electrode potential of the following cell at 298K. Given $E^{\circ}_{\text{cell}}=1.3\text{V}$.

Cell representation $\text{Cu}_{(\text{s})} / \text{Cu}^{2+}_{(1 \times 10^{-2} \text{ M})} // \text{Ag}^{+}_{(1 \times 10^{-1} \text{ M})} / \text{Ag}_{(\text{s})}$

Electrode reactions are

 E_{cell}

$$E_{\text{Cell}} = E^{\circ}_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$$

$$E_{\text{Cell}} = 1.3 - \frac{0.0591}{2} \log \frac{[1 \times 10^{-2}]}{[1 \times 10^{-1}]^2}$$

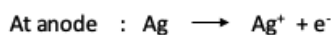
$$E_{\text{cell}} = 1.3 - 0.0591/2 \{ \log 10^{-2} - \log 10^{-2} \}$$

$$E_{\text{cell}} = 1.3\text{V}$$

11. Calculate the electrode potential of the following concentration cell at 298K.

Cell representation $\text{Ag}_{(\text{s})} / \text{Ag}^{+}_{(0.05\text{M})} // \text{Ag}^{+}_{(0.5\text{M})} / \text{Ag}_{(\text{s})}$

Electrode reactions are



$$E_{\text{Cell}} = \frac{0.0591}{n} \log \frac{[M_2]}{[M_1]} \quad \text{Where } M_2 > M_1$$

$$E_{\text{Cell}} = \frac{0.0591}{1} \log \frac{[0.5]}{[0.05]}$$

$$E_{\text{cell}} = 0.0591 \text{ V}$$

12. Calculate the pH of a solution in which a glass electrode is placed and connected with a standard calomel electrode. The emf of the cell is found to be 0.034V. Standard calomel electrode potential is 0.2422V and $E^{\circ}_{\text{G}}=0.8304\text{V}$.

$$\text{pH} = \frac{E^{\circ}_{\text{G}} - E_{\text{cal}} - E_{\text{cell}}}{0.0591}$$

$$\text{pH} = \frac{0.8304 - 0.2422 - 0.034}{0.0591}$$

$$= 9.3$$

13. Two silver electrodes containing 0.0012M and 0.0137M AgNO_3 solution respectively are coupled by salt bridge. Formulate the concentration cell. Calculate the valency of silver ions if the emf of the cell is 0.0627V at 298 K

Representation of the cell $\text{Ag}_{(\text{s})} / \text{AgNO}_3_{(0.0012\text{M})} // \text{AgNO}_3_{(0.0137\text{M})} / \text{Ag}_{(\text{s})}$

 E_{cell}

$$E_{\text{Cell}} = \frac{2.303RT}{nF} \log \frac{[M_2]}{[M_1]} \quad \text{Where } M_2 > M_1$$

Here, $M_2=0.0137 \text{ M}$; $M_1=0.0012 \text{ M}$; $T=298 \text{ K}$; $R=8.314\text{J/K}$; $F=96500$; $E_{\text{cell}} = 0.0627\text{V}$

$$n = \frac{2.303 \times 8.31 \times 298 \log \frac{[0.0137]}{[0.0012]}}{0.0627 \times 96500}$$

$$= 0.9972$$

$$= 1 \quad \text{Silver in } \text{AgNO}_3 \text{ is univalent.}$$

14. Calculate the electrode potential when copper electrode is dipped in a 0.125M CuSO₄ solution which is 80% dissociated into copper ions

$$E^0_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$$

$$E_{\text{Cu}^{2+}/\text{Cu}} = E^0_{\text{Cu}^{2+}/\text{Cu}} + \frac{0.0591}{n} \log [\text{Cu}^{2+}]$$

$$[M^{n+}] = \frac{0.125 \times 80}{100} = 0.1$$

$$E_{\text{Cu}^{2+}/\text{Cu}} = 0.34 + \frac{0.0591}{2} \log [0.1] = 0.31\text{V}$$

15. Two copper electrodes placed in copper sulphate solutions of equal concentration are connected to form a concentration cell. What is the cell voltage if one of the solution is diluted until the concentration of Cu²⁺ ions is 1/5th of its original value. What is the cell voltage after dilution?

$$E_{\text{Cell}} = \frac{0.0591}{n} \log \frac{[M_2]}{[M_1]} \quad \text{Where } M_2 > M_1$$

$$E_{\text{Cell}} = \frac{0.0591}{2} \log \frac{[1]}{[1/5]}$$

$$E_{\text{Cell}} = \frac{0.0591}{2} \log 5 = 0.0206\text{V}$$

16. Calculate the cell potential of Ag/Ag⁺ coupled with Cu/Cu²⁺ if the concentration of Ag⁺ and Cu²⁺ are 4.2x10⁻⁶M and 1.3x 10⁻³M respectively. E⁰_{Cell} = 0.46 V. what is the value of delta G for the reduction of 1 mole Ag⁺. Given F=96.5k J V⁻¹mol⁻¹ (KJ per volt gram equivalent)

$$E_{\text{Cell}} = E^0_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[\text{Reduced form}]}{[\text{Oxidized form}]}$$

$$E_{\text{Cell}} = E^0_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[\text{Ag}^+]^2}{[\text{Cu}^{2+}]}$$

$$E_{\text{Cell}} = 0.46 - \frac{0.0591}{2} \log \frac{[4.2 \times 10^{-6}]^2}{[1.3 \times 10^{-3}]} = 0.23 \text{ V.}$$

For the reduction of 2 mole of Ag⁺ ions 2 equivalent required

For the reduction of 1 mole of Ag⁺ ions 1 equivalent required

Change in free energy for the reduction 1 mole of Ag⁺

$$\Delta G = -nFE$$

$$= -1 \times 96.5 \times 0.23$$

$$\Delta G = -22 \text{ KJ}$$